

Network Defense

Firewalls and Tunnels

BCS2420

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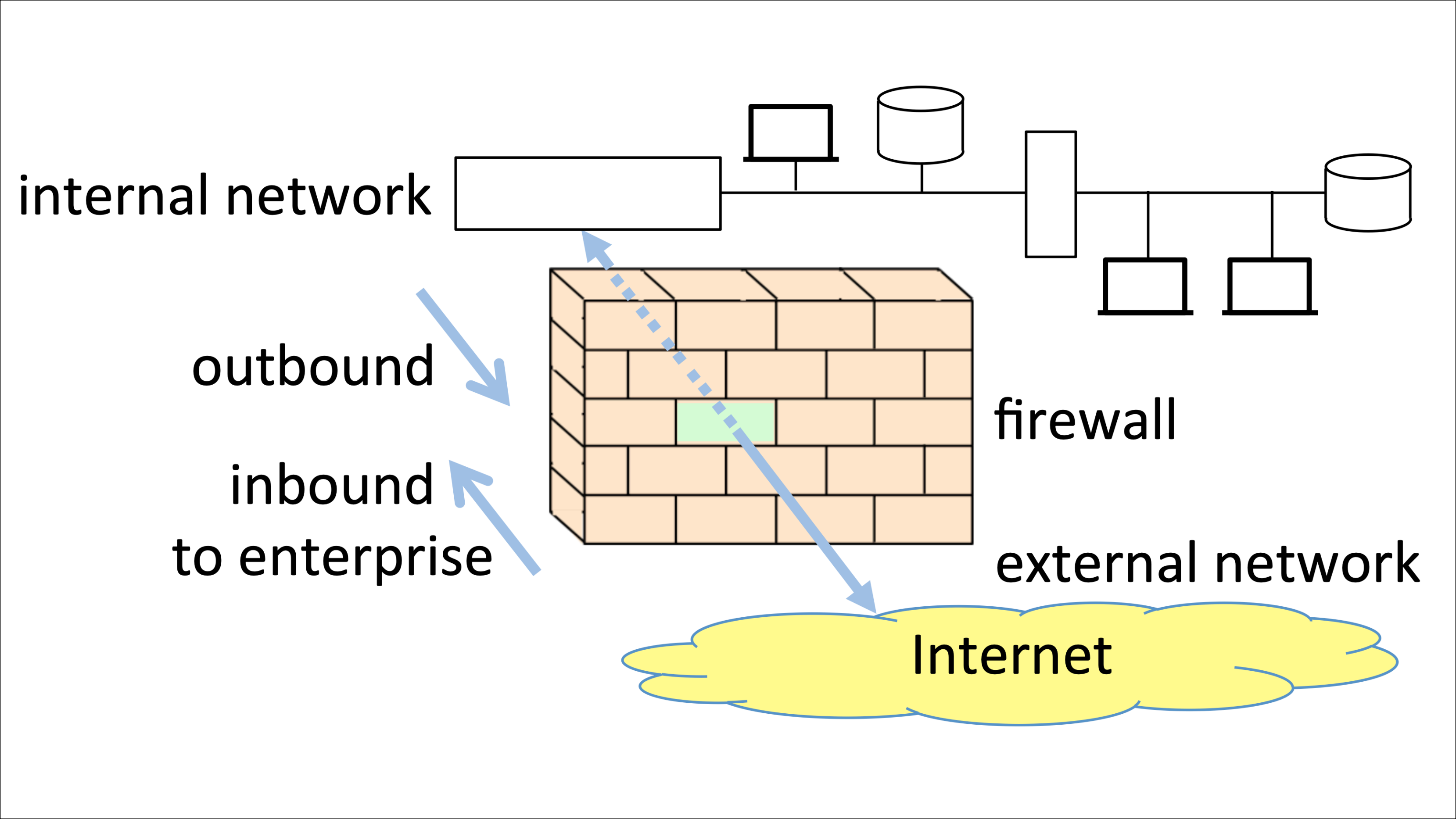
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Introduction

Firewalls and encrypted tunnels (VPNs) provide perimeter-based defenses.

Firewalls Overview

- Firewalls serve as gateways, controlling access between *networks* or *devices*, isolating damage, and containing spread using principles like Complete-mediation And Isolated-compartments.
- Often used in perimeter defenses, they protect trusted internal networks from untrusted external ones (e.g., the Internet).



Packet-Filter Firewalls

A packet-filter firewall uses rules to allow or deny data packets based on header fields. Rules follow a

`<condition, action>` format, with actions like `ALLOW`, `DROP`, or `REJECT`.

Stateless vs. Stateful Filters

- **Stateless Filters:** Process each packet independently.
- **Stateful Filters:** Keep track of connection states, processing packets based on prior state information.

Packet-Filter Rules and Actions

Example Rules

1. **Ingress/Egress Filtering:** Rules to prevent spoofed IP addresses.
2. **SMTP Email:** Rules to allow or deny email traffic based on IP addresses and ports.
3. **HTTP Traffic:** Rules to manage outbound and inbound `HTTP` connections, using flags like `ACK`.
4. **DNS Queries:** Rules to control DNS traffic for internal and external queries.

Default-Deny Rulesets

Principle of SAFE-DEFAULTS

A default-deny ruleset **blocks** all packets unless **explicitly allowed** by **an accept rule**, reducing the risk of unknown exploits and aligning with strict security policies.

Firewall and Security Policy

Firewalls **enforce** an organization's **Internet security policy** by defining rules that **allow** or **block** packets, translating authorized services into specific access controls.

Firewalls as Chokepoints

- Firewalls serve as **centralized points** for *monitoring*, *control*, and *packet rejection*, assuming a secure perimeter. Despite the shift from single-point network access, they remain valuable for:
 - Protecting legacy applications.
 - Enforcing security policies against remote threats.
 - Supporting defensive strategies like defense-in-depth and isolation.

Limitations of Firewalls

1. **Topological Limitations:** Assumes true perimeters exist.
2. **Malicious Insiders:** Provides little protection against trusted users cooperating with outsiders.
3. **Bad Connections by Trusted Users:** Limited protection against malicious content from compromised sites.
4. **Tunneling:** Firewall rules can be bypassed by tunneling disallowed protocols.
5. **Encrypted Content:** Prevents content-based inspection unless decrypted at the firewall.

Dynamic Packet Filtering Example: FTP ¹

FTP uses separate **TCP** connections for commands and data. Dynamic packet filters track **client-specified ports** to temporarily **allow** inbound connections, resolving firewall conflicts.

1. FTP (File Transfer Protocol) is a standard network protocol used to transfer files between a client and server over TCP.

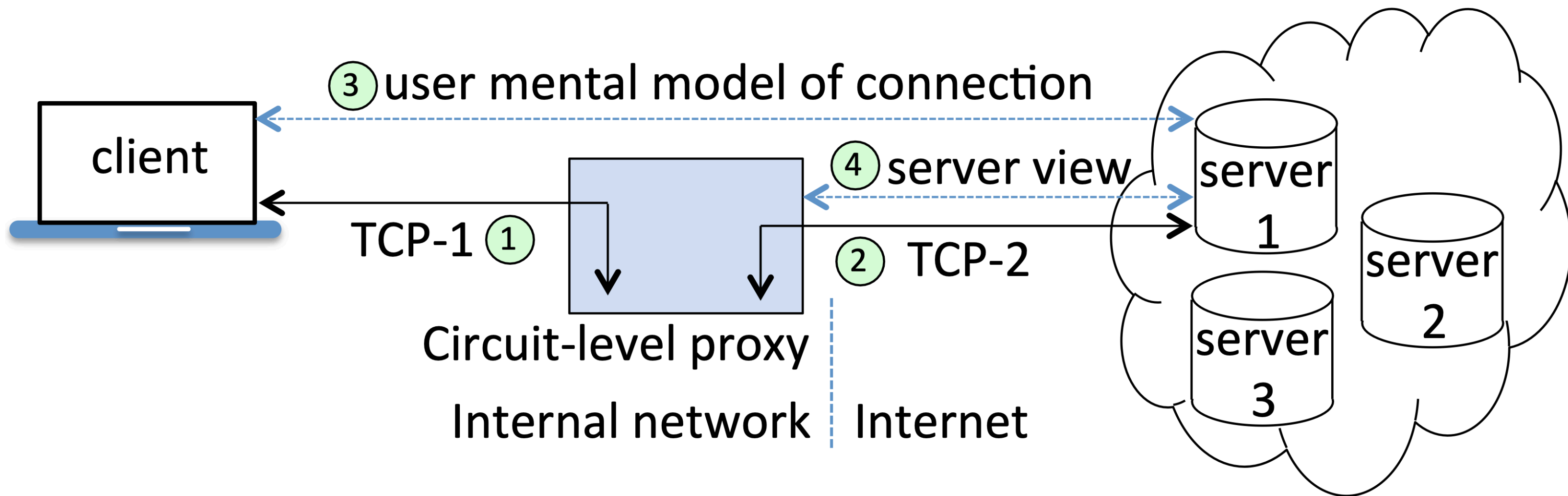
Proxy Firewalls and Firewall Architectures

Proxy Firewalls

Act as intermediaries between **internal clients** and **external services**, offering **circuit-level** and **application-level** filtering.

Circuit-Level Proxies

Relays connections through a proxy point, allowing or denying the connection and then relaying data. **SOCKS** is a common protocol used.



Application-Level Filters

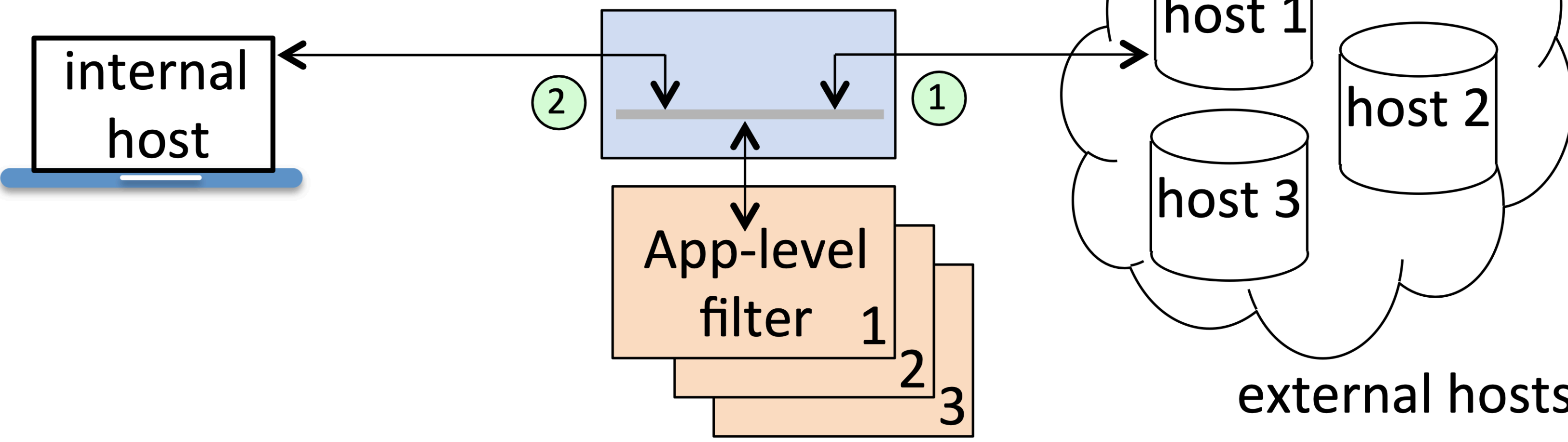
Uses specialized *programs* to filter traffic based on application-specific protocols, capable of altering payloads for intrusion prevention.

Targeted Applications

Includes widely-used protocols like TELNET, FTP, HTTP, and email, which are filtered for security issues.

Internal network : Internet

Application-level gateway :



Bastion Hosts and Dual-Homed Hosts

Bastion Hosts

Exposed to **hostile networks** and hardened by **disabling non-essential services**.

Dual-Homed Hosts

Computers with **two** network interfaces used as part of **multi-component firewalls** to ensure no direct connections between external and internal networks.

Enterprise Firewall Architectures

Single Screening Router

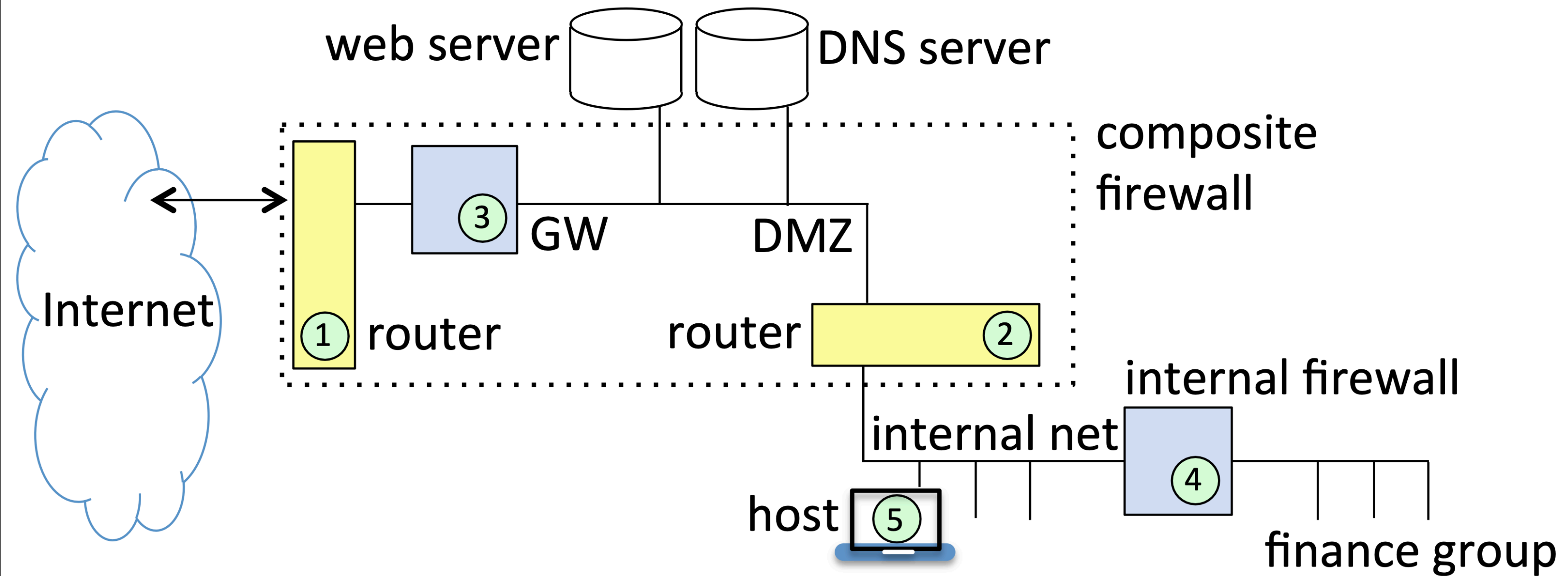
Basic protection but **limited** configurability.

Screening Router and Bastion Host

Provides additional security by having a **bastion host** behind the router.

DMZ (Demilitarized Zone)

A **subnetwork** between **external** and **internal** networks, using **multiple screening routers** and a bastion host for added security.



Secure Shell (SSH)

Secure Shell (SSH)

SSH enables secure, encrypted communication, replacing insecure protocols like telnet and FTP. It consists of:

1. **Transport Layer:** Ensures encryption and integrity.
2. **Authentication:** Manages client-server authentication.
3. **Connection:** Allows multiple sessions over one connection.

TARGET	FULL NAME	FUNCTIONALITY
rsh (ssh)	remote shell	Send shell commands for execution on remote host by a daemon (rshd).
rlogin (ssh)	remote login	Log in to remote Unix server over TCP network, then communicate as if physically local.
telnet (ssh)	teletype network	Acquire interactive virtual terminal connection over TCP, e.g., to a command line interface (Unix, Windows).
ftp (sftp)	file transfer prot (secure ftp)	Transfer files, using separate connections for control and data. Can also be replaced by ftps (FTP over TLS).
rcp (scp)	remote copy (secure copy)	Copy files, directories between local, remote systems, with command line syntax similar to Unix command cp

local
host

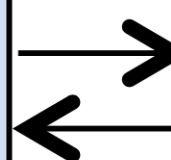
remote
host

SSH
client

SSH tunnel

SSH
server

shell (cmd
interpreter)



SSH Client Authentication

Methods

1. **Client Password**
2. **Kerberos Ticket**
3. **Client Public Key**

Process

1. Client sends **public key** and **signature**.
2. Server verifies **key** and **signature**.

SSH Server Authentication

Establishing Trust

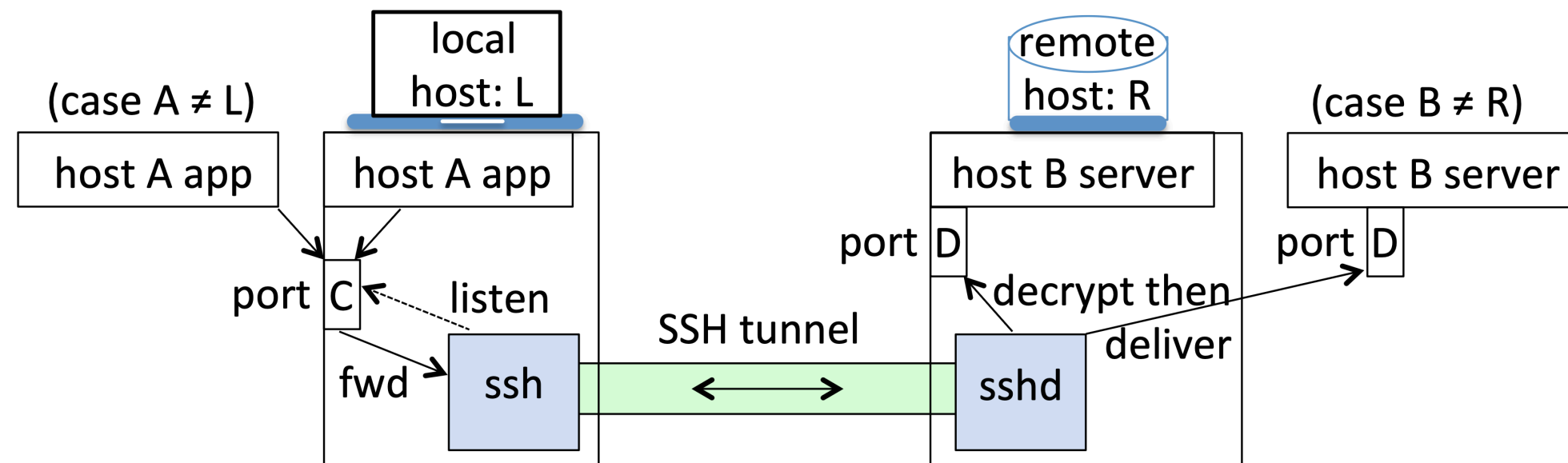
Server authentication involves a **public key** (SSH host key) verified by the client. Trust models include:

1. **Client Database of Server Keys:** Trusted keys stored locally.
2. **CA-Certified Server Keys:** Verified using a CA's **public key**.

SSH Port Forwarding

Local Port Forwarding

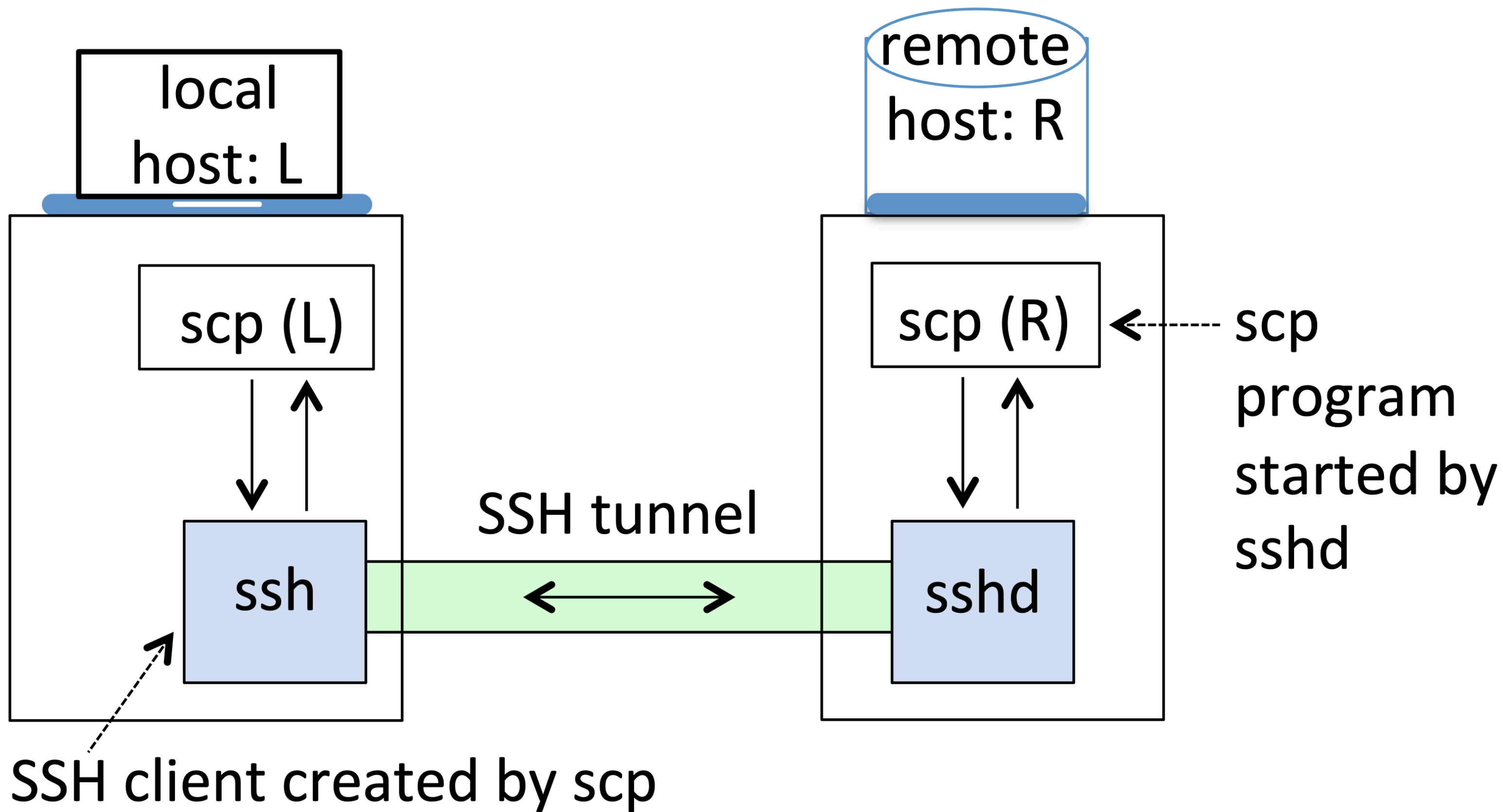
Redirects data from an **unsecured application** through an **SSH tunnel**.



SCP (Secure Copy)

Functionality

Transfers files securely using `SSH` tunnels, replacing `r``cp`.



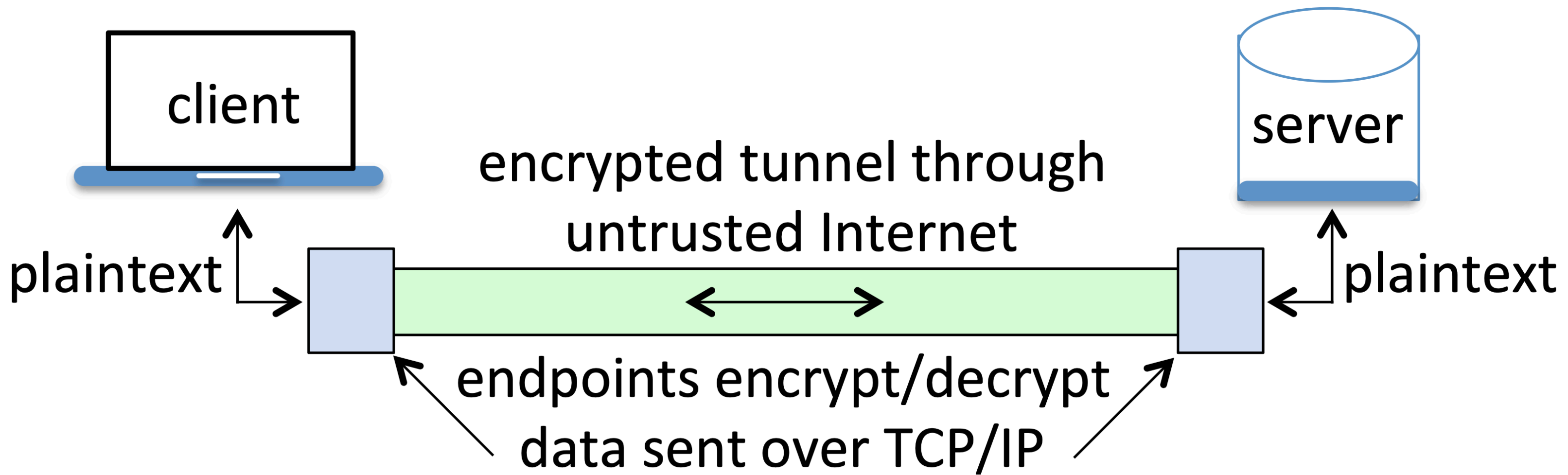
VPNs and Encrypted Tunnels

Motivation

Encrypts data to protect against **eavesdropping** and **tampering**.

Tunneling

Encapsulates **one protocol** within **another** for secure transit.



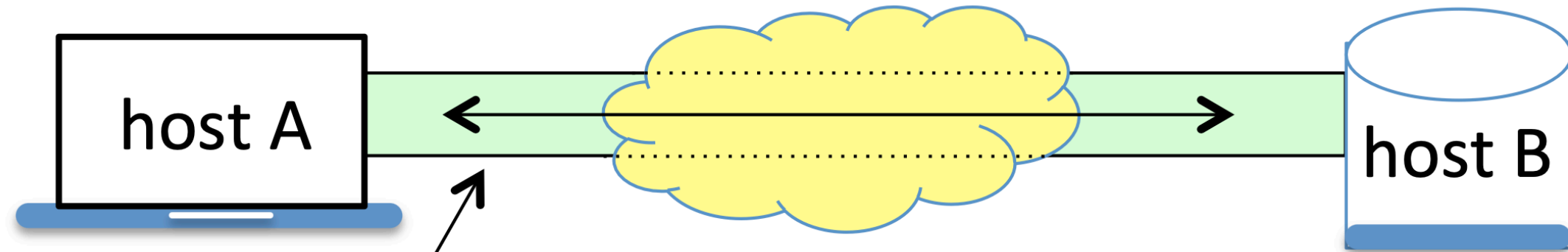
Virtual Private Networks (VPNs)

A private network using **encrypted tunnels** over **public networks**.

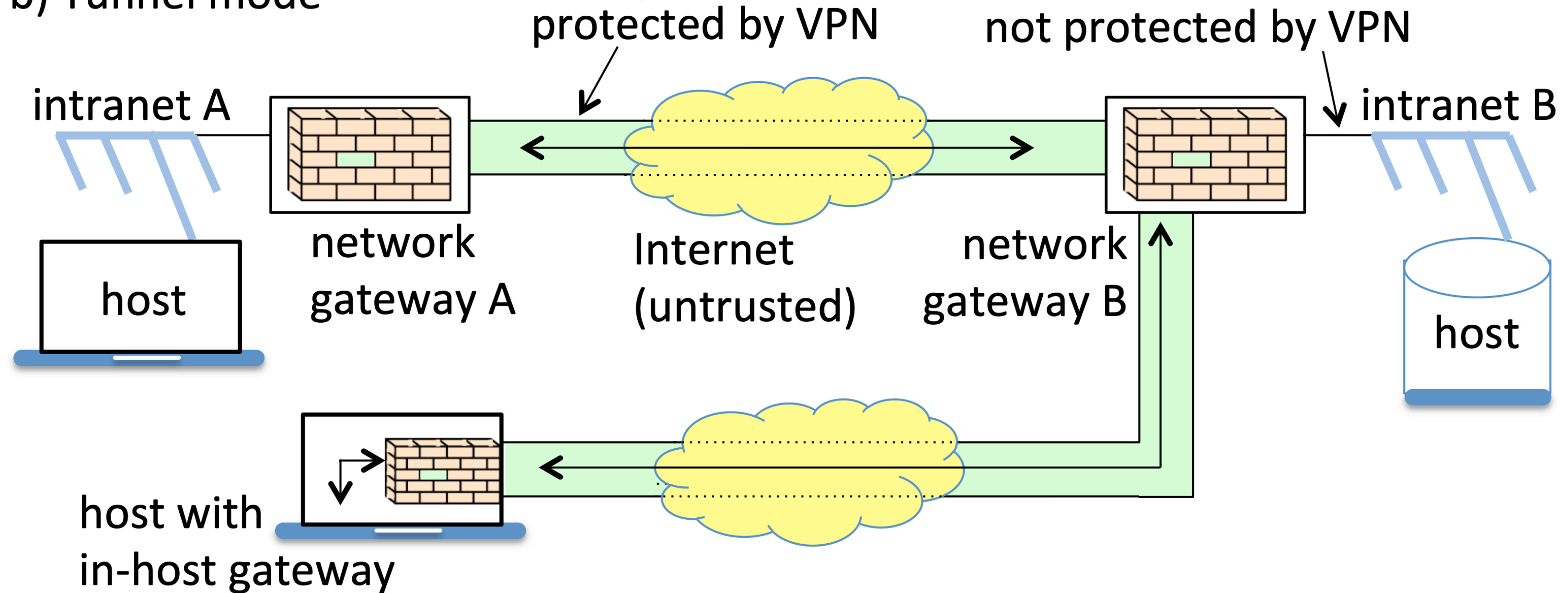
Designs and Use Cases

1. **Transport Mode:** Host-to-host VPN for end-to-end security.
2. **Tunnel Mode:** Network-to-network or host-to-network VPNs for site-to-site or remote access.

a) Transport mode



b) Tunnel mode



Application Protocol (HTTP, FTP, DNS)		SSH
TLS	Transport (TCP, UDP)	
Network (IP, ICMP)		IPsec
Link (Ethernet, Wi-Fi, ARP)		

